Solution Architecture

Date	15 March 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyzer
Maximum Marks	5 Marks

Solution Architecture Diagram

The Voice-Based Concept Understanding Analyzer follows a layered architecture consisting of Presentation, Application and Data layers. Audio is uploaded through the client interface, processed by AI services, and stored securely along with generated reports.

Presentation / Client Layer

Streamlit Web Application
Responsive User Interface

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API Gateway / Server

Python • Streamlit Server

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Authentication

User Login
Session Management

Core Logic

Speech-to-Text
Semantic Analysis
Audio Feature Extraction
PDF Report Generation

External APIs

OpenAI Whisper
Sentence Transformers
Librosa

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Data / Storage Layer

JSON Files • Reports • Audio Files

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for audio upload, report viewing and interaction.	HTML5, CSS3, Streamlit
API Gateway	Routes requests between the client interface and AI services.	Python, Streamlit
Core Logic Service	Performs transcription, semantic similarity analysis, scoring and report generation.	OpenAI Whisper, Sentence Transformers, Librosa, FPDF
Database / Storage	Stores reference concepts, generated reports and uploaded audio.	JSON, Local Storage